

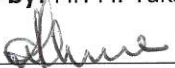
BIIM3110 – INTERACTIVE MULTIMEDIA

COURSE DETAILS:

Faculty:	Information and Communication Technology
Programme:	BSCSM - Degree in Software Engineering with Multimedia BSCIT - Degree in Information Technology
Year/Semester:	Year 3, Semester 2
Lecturer(s):	Mr M. Takaimane & Mr. T.Thokoana
Principal Lecturer:	Ms M. Nthunya
Pre-requisite:	OOP2
Methods of Delivery:	Lecture
Credits:	12
Contacts hours:	3 hours (2 lecture +1 tutorial)
Number of weeks:	17 weeks long semester (12 teaching, 1 midterm examination, 1 non-contact 1 revision, 4 Assessments)
No. of assessments	4
Venue:	MM7
Day:	Thursday
Time:	14:30 – 16:30

Prepared by: Mr. M. Takaimane

Checked by: Mr. M. Nthunya

Signature:  Date: 30/01/2026

Signature:  Date: 02/02/26

This document comprises the following:

- Essential information
- Specific course information
- Course rules & regulations
- Grades
- Plagiarism
- Course introduction
- Course aims & objectives
- Learning outcome
- Specific generic learning skills
- Syllabus + lecture outline
- References
- Assignment schedule
- Assessment criteria
- Specific criteria

Other documents as follows will be issued to you on an ongoing basis throughout the semester:

- Handouts for lecture notes and tutorials
- Submission Requirements + Guideline

1.0 ESSENTIAL INFORMATION

- All courses other than electives are '**significant Courses**'
- As an indicator of workload, one (1) credit carries and additional two (2) hours of self-study per week. For example, a course worth 10 credits require that the student spends an additional 20 hours in a semester through reading, completing the assessment or doing self-directed research for that course.
- Writing and submission of **ALL** assessments work is compulsory in this Course. A student cannot pass this Course without having to submit **ALL** assignment work by the due date or an approved extension of that date.
- All assignments are to be handed on time on the due date. Students will be penalised 10 percent for the first day and 5 percent per day thereafter for late submission (a weekend or a public holiday count as one day). Late submission, after the date Board of Studies meeting will not be accepted.
- Due dates, compulsory assignment requirements and submission requirements may only be altered with the consent of the majority of students enrolled in this Course at the beginning/early in the program.
- Extensions of time for submission of assignment work may be granted if the application for extension is accompanied by a medical certificate.
- Overseas travel is not an acceptable reason for seeking a change in the examination schedule.
- Only the Head of School can grant approval for extension of submission beyond the assignment deadline.
- Re-submission of work can only receive a 50% maximum pass rate.
- Supplementary exams can only be granted if the level of work is satisfactory **AND** the semester work has been completed.
- Harvard referencing and plagiarism policy will apply on all written assignments.

2.0 SPECIFIC COURSE INFORMATION

- Attendance rate of 80% is mandatory for passing Course.
- All grades are subject to attendance and participation.
- Absenteeism at any scheduled presentations will result in zero mark for that presentation.
- Visual presentation work in drawn and model form must be the original work of the student.
- The attached semester program is subject to change at short notice.

3.0 COURSE RULES AND REGULATIONS:

Assessment procedure:

- These rules and regulations are to be read in conjunction with the **Unit Aims and Objectives**
- All assignments must be completed and presented for marking by the due date.
- Marks will be deducted for late work and invalid reasons.
- All assignments must be delivered by the student in person to the lecturer concerned. No other lecturer is allowed to accept students' assignments.
- Assignments may be submitted through the designated learning management system in place.
- All tests/examinations are compulsory.
- Students must sit for the test/examination on the notified date.
- Students are expected to familiarize themselves with the test/examination timetable.
- Students who miss a test/examination will not be allowed to pass.
- Any scheduling of tutorials, both during or after lecture hours, is **TOTALLY** the responsibility of each student. Appointments are to be proposed, arranged, confirmed, and kept, by each student. Failure to do so in a professional manner may result in penalty of grades. Tutorials **WITHOUT** appointments will also **NOT** be entertained.
- Note that every assignment is given an ample time frame for completion. This, together with advanced information pertaining deadlines gives you **NO EXCUSE** not to submit assignments on time.

4.0 GRADES

All Courses and assessable projects will be graded according to the following system. With respect to those units that are designated 'Approved for Pass/Fail' the grade will be either PX or F:

Grade	Numeric	Grade description
90 – 100	A+	
85 – 89	A	Pass with Distinction
80 – 84	A-	
75 – 79	B+	
74 – 70	B	Pass with Credit
65 – 69	B-	
60 – 64	C+	
55 – 59	C	
50 – 54	C-	Pass
50	PX	Pass after extra work
0 – 49	F	
	EXP	Exempted
	PP	Pass Provisional with extra work needed
	PX	Pass after extra work is given and passed
	X	Ineligible for assessment due to unsatisfactory attendance
	DEF	Deferred
	W	Withdraw
	DNA	Did Not Attend Course
	DNC	Did Not Complete Course

5.0 PLAGIARISM, COPYRIGHT, PATENTS, OWNERSHIP OF WORK: STUDENT MAJOR PROJECT, THESES & WORKS

5.0.1 Academic Dishonesty

Academic Dishonesty or academic misconduct is any type of cheating that occurs in relation to a formal academic exercise. It can include:

Plagiarism: The adoption or reproduction of original creations of another author (person, collective, organization, community or other type of author, including anonymous authors) without due acknowledgement.

Fabrication: The falsification of data, information or citations in any formal academic exercise.

Deception: Providing false information to an instructor concerning a formal academic exercise – e.g., giving a false excuse for missing a deadline or falsely claiming to have submitted work

Cheating: Any attempt to give obtain assistance in a formal academic exercise (like an examination) without due acknowledgement.

Bribery: or paid services. Giving certain test answers for money.

Sabotage: Acting to prevent others from completing their work. This includes cutting pages out of library books or wilfully disrupting the experiments of others.

Students involved in academic dishonesty will be treated as follows:

1. Plagiarism and fabrication in Assignments, Projects or Presentations: A zero (0) mark will be awarded to the work.
2. **Cheating:** Any form of assistance in a test will be rewarded a zero (0) mark. While in an examination the student automatically repeats the module.
3. **Bribery, Sabotage or Professional Misconduct:** Committing any of these offenses will lead to discontinuity from the University for a certain period depending on the seriousness of the offence.

5.0.2 Copyright, Patents and Ownership of Work

All use of the borrowed work, whether text or images must be referenced and acknowledged. Limkokwing is not liable for any infringement of copyright within students' work. Students are to abide by International Copyright laws. Students engaged in a project activity will be required, as a condition of acceptance as a student, to agree to assign to Limkokwing their right, title and interest in any invention or visual work arising from their studies with the University. Limkokwing has the right to keep students work for use in archives, exhibitions and events. Where students wish to keep original work, they must apply in writing and gain permission from the Academic Management.

6.0 COURSE INTRODUCTION

This course is designed to provide students with the knowledge and skills necessary to develop rich and interactive multimedia applications using JavaFX. Students will learn how to build user-friendly graphical user interfaces (GUIs), and implement multimedia features such as audio, video, and animations. By the end of the course, students will have developed a solid foundation in JavaFX and be able to create professional-looking multimedia applications.

7.0 COURSE AIMS AND OBJECTIVES

1. Introduce students to the concepts and principles of interactive multimedia development using JavaFX.
2. Familiarize students with the JavaFX framework and its tools for creating multimedia applications.
3. Provide hands-on experience in designing and implementing interactive multimedia applications, to meet the demands of the rapidly changing industry.
4. Develop students' proficiency in media handling, including image and audio/video playback.
5. Enable students to apply styling and customization techniques to enhance the visual appeal of multimedia applications.
6. Cultivate problem-solving and critical thinking skills specific to interactive multimedia development.

8.0 LEARNING OUTCOME

Students will be able to:

1. Understand the fundamental concepts and principles of interactive multimedia development using JavaFX.
2. Demonstrate proficiency in media handling, including image loading, audio/video playback, and manipulation.
3. Employ animation and interactivity to create engaging user experiences to fill the increasing demand of multimedia applications
4. Understand the basics of 3D graphics and utilize predefined 3D shapes and transformations in multimedia development.

9.0 SPECIFIC GENERIC LEARNING SKILLS

Upon completion of the Course, student will acquire skills in:

- Problem Solving: Students will develop the ability to analyze complex problems, identify potential solutions, and select and implement the most appropriate solution using JavaFX.
- Design and Creativity: Students will learn how to design and create visually appealing and interactive user interfaces using JavaFX, incorporating multimedia features such as images, audio, and video.
- Programming: Students will develop a strong foundation in Java programming and object-oriented programming concepts, techniques, and best practices.

- Analytical Thinking: Students will learn to evaluate and optimize the performance of JavaFX applications through analyzing, debugging, and testing.
- Communication: Students will develop effective communication skills through team-based projects, presentations, and discussions.
- Self-Directed Learning: Students will develop the ability to learn independently, take responsibility for their own learning, and seek out resources and information to solve problems and complete projects.

10.0 PROGRESSION POLICY

Students can only enroll into a higher semester if they have passed all courses in the preceding semester or have failed not more than two courses. A student cannot register for any course at a higher level if they fail its lower-level unit which in essence is a pre-requisite: If student CGPA is less than 2.0 could not proceed to the next semester.

11.0 UNIT SYLLABUS + LECTURE OUTLINE

Week	1
LECTURE	1: Getting Started with JavaFX
Lecture Synopsis:	• What Is JavaFX • Setting-up the Environment • Your First JavaFX Application • Layouts and Controls
Handout:	Course Outline

Week	2
LECTURE	2: Managing Stages and Scenes
Lecture Synopsis:	• What is a Stage • Showing the Primary Stage • Setting the bounds of a Stage • Resizing a Stage • Graphic Rendering Modes • Setting the Cursor for a Stage • Understanding the Platform Class
	Assignment 1

Week	3
LECTURE	3: Understanding Nodes
Lecture Synopsis:	• The Cartesian Coordinate System • Knowing the Bounds of a Node • Bounds of a Group • Positioning a Node using layout • Sizing Nodes • Understanding Colors • Linear Color Gradient • Radial Color Gradient

Week	4
LECTURE	4. Styling Nodes
Lecture Synopsis:	• Adding Style Sheets • Default Style Sheet • Default Style Sheet • Priorities of Styles for a Node • Types of CSS Properties • Understanding Style Selectors
Handout:	Assignment 2
Submission	Assignment 1

Week	5
LECTURE	5: Image API
Lecture Synopsis:	• What Is the Image API • Loading an Image • Viewing an Image • Understanding Image Operations
Submission	Assignment 2

Week	6
LECTURE	6: Playing Audios and Videos
Lecture Synopsis:	• Understanding the Media API • Playing Short Audio Clips • Playing Media
Handout:	Assignment 3

week 7:	Mid-Term Examinations
week 8:	[semester break]

Week	9
LECTURE	7: Understanding 2D Shapes
Lecture Synopsis:	• Drawing 2D Shapes • Building Complex Shapes Using the Path Class • Drawing Scalable Vector Graphics
Handout:	Group Project, Assignment 4
Submission	Assignment 3

Week	10
LECTURE	8: Transformations
Lecture Synopsis:	• The Translation Transformation • The Rotation Transformation • The Scale Transformation • The Shear Transformation

Week	11
LECTURE	9: Animation
Lecture Synopsis:	• Understanding Animation Classes • Utility Classes • Timeline Animation • Controlling Animation • Translation • Interpolators

Week	12
LECTURE	10: Drawing on Canvas
Lecture Synopsis:	• What is the Canvas API • Creating a Canvas • Test, Paths, Pixels and Images • Saving and Restoring Drawing States
Submission:	Assignment 4

Week	13
LECTURE	11: Case Study
Lecture Synopsis:	• A 2D Game
Summation	Group Project

Week	14
LECTURE	12: Understanding 3D Shapes
Lecture Synopsis:	• The 3D Coordinate System • Using Predefined 3D Shapes • Using Cameras • Using Light Sources
Handout:	

WEEK	15	REVISION
WEEK	16	FINAL EXAMINATIONS
WEEK	17	FINAL EXAMINATIONS

12.0 REFERENCES

Recommended Books:
Learn JavaFX 17: Building User Experience and Interfaces with Java, Kishori Sharan, 2022.
Additional Books:
JavaFX: Developing Rich Internet Applications by Jim Clarke, Jim Connors, and Eric J. Bruno, 2014

13.0 ASSESSMENT SCHEDULE

Assessment description	Issue date	Due date	%
Assignment 1	Week 1	Week 3	10
Assignment 2	Week 3	Week 5	20
Assignment 3	Week 6	Week 9	20
Assignment 4	Week 9	Week 12	25
Group Project	Week 9	Week 13	25
Total			100

14.0 ASSESSMENT CRITERIA

You will be graded based on your effort in submitting successfully well worked out assignments, projects, tests, and performance in the examinations.

15.0 LEARNING ACTIVITIES

- In-class coding challenges and exercises to reinforce concepts and practice problem-solving skills
- Group projects to build real-world applications and collaborate with classmates
- Pair programming sessions to work on challenging tasks and learn from each other
- Use of online resources such as tutorials, forums, and documentation to expand knowledge and skills.
- Independent research projects where students delve deeper into specific topics and present their findings to the class

16.0 SPECIFIC CRITERIA

- Each assignment will be handed out with the project brief and will vary, depending on the teaching and learning objectives of the specific assignment.
- Each student will receive a completed assessment sheet back with their marks, thereby giving student feedback on each set criterion and the project as a whole.
- Feedback for each assessment will be issued out within 2 weeks of hand-in date.

